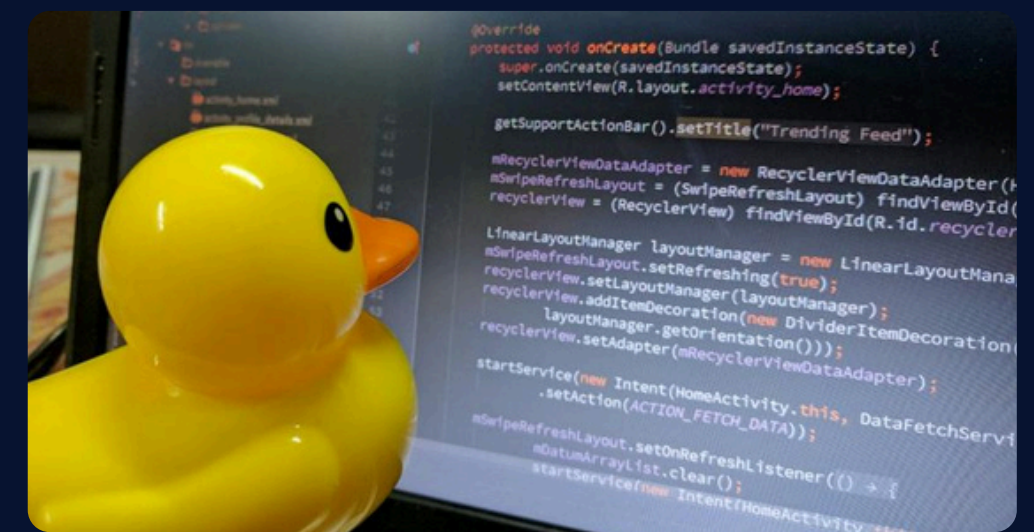




Logistic Regression



Session -> 6



Intro to Logistic Regression

Note

- “Logistic Regression” has “Regression” in its name, **but** it is a classification algorithm, not regression.

Machine Learning Types

```
graph TD; A[Machine Learning Types] --> B[Supervised Learning]; A --> C[Unsupervised Learning]; A --> D[Reinforcement Learning]; B --> E[Resgression]; B --> F[Classification]; C --> G[Clustering]; C --> H[Dimensionality Reduction]; style F fill:#192333,color:#fff
```

Supervised Learning

Unsupervised Learning

Reinforcement Learning

Resgression

Classification

Clustering

**Dimensionality
Reduction**

What is Classification?

- The goal of classification is to predict which class an input belongs to among the existing classes.

Example: Diabetes Prediction

Goal → predict whether a patient is likely to have diabetes based on their health data.

Classes(Target) → Yes (1) , No (0)

Age	Blood Pressure	Diabetes
45	85	Yes
28	75	No
50	88	Yes

Binary **VS** Multiclass Classification

- **Binary Classification:** Two possible outcomes (e.g., Yes/No).

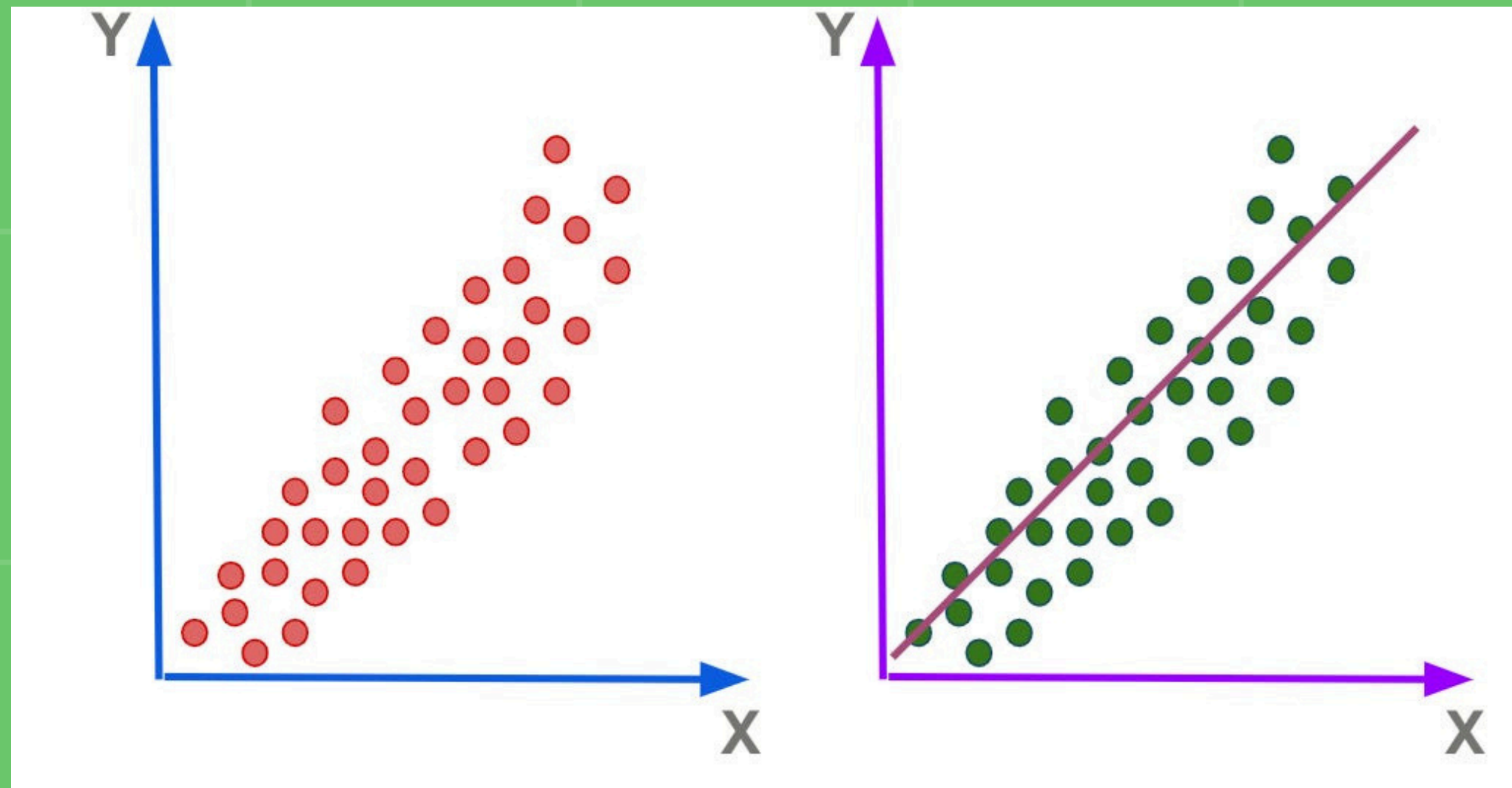
Example: Coronavirus Prediction

- **Multiclass Classification:** More than two classes (e.g., Cat/Dog/Bird).

Example: Disease Type Classification

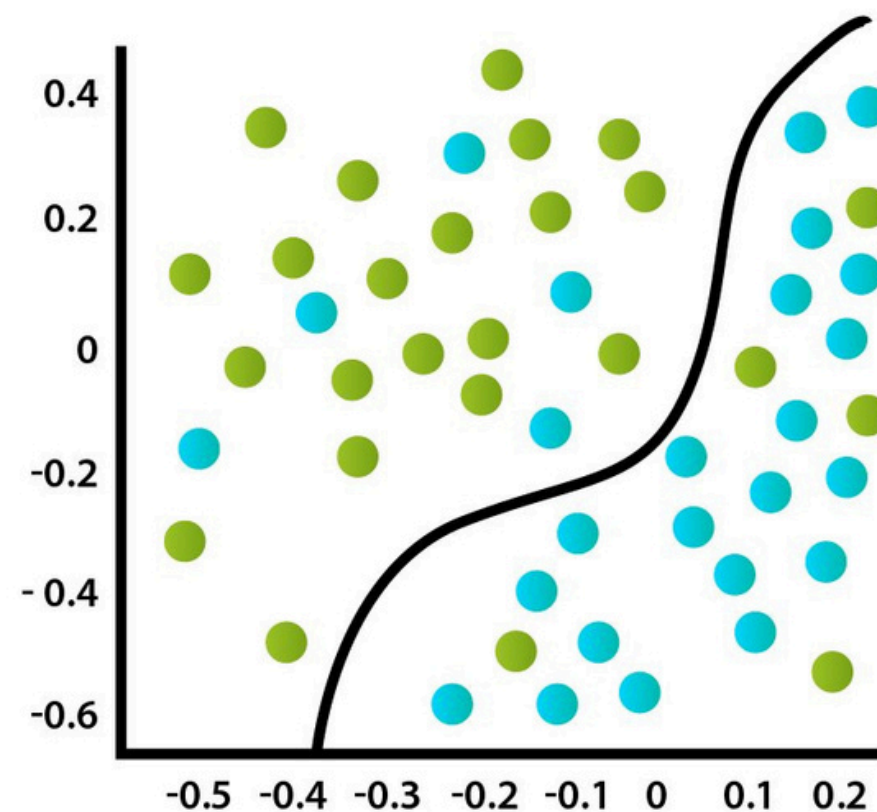
Reminder of Linear Regression

- Remember in Linear regression, our goal was to find the line that can fit the dataset, so it can predict values as close as possible to the target(minimize the cost).

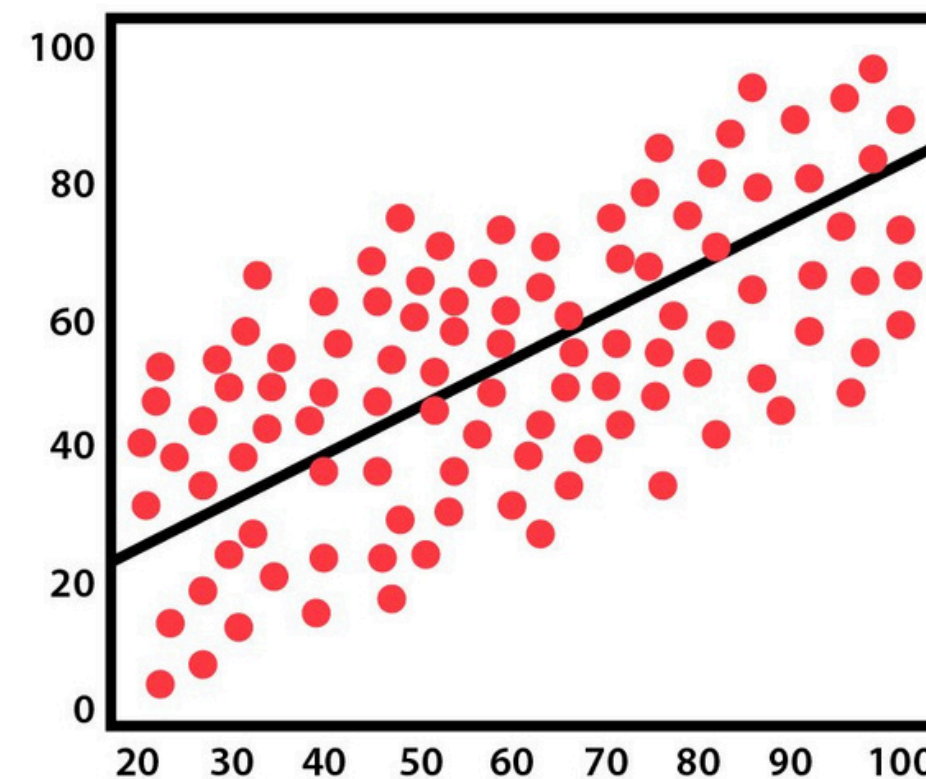


What about Logistic Regression?

- In Logistic regression, we want to separate the datapoints into two classes (binary classification), so we want a line that can distinguish each class.



CLASSIFICATION



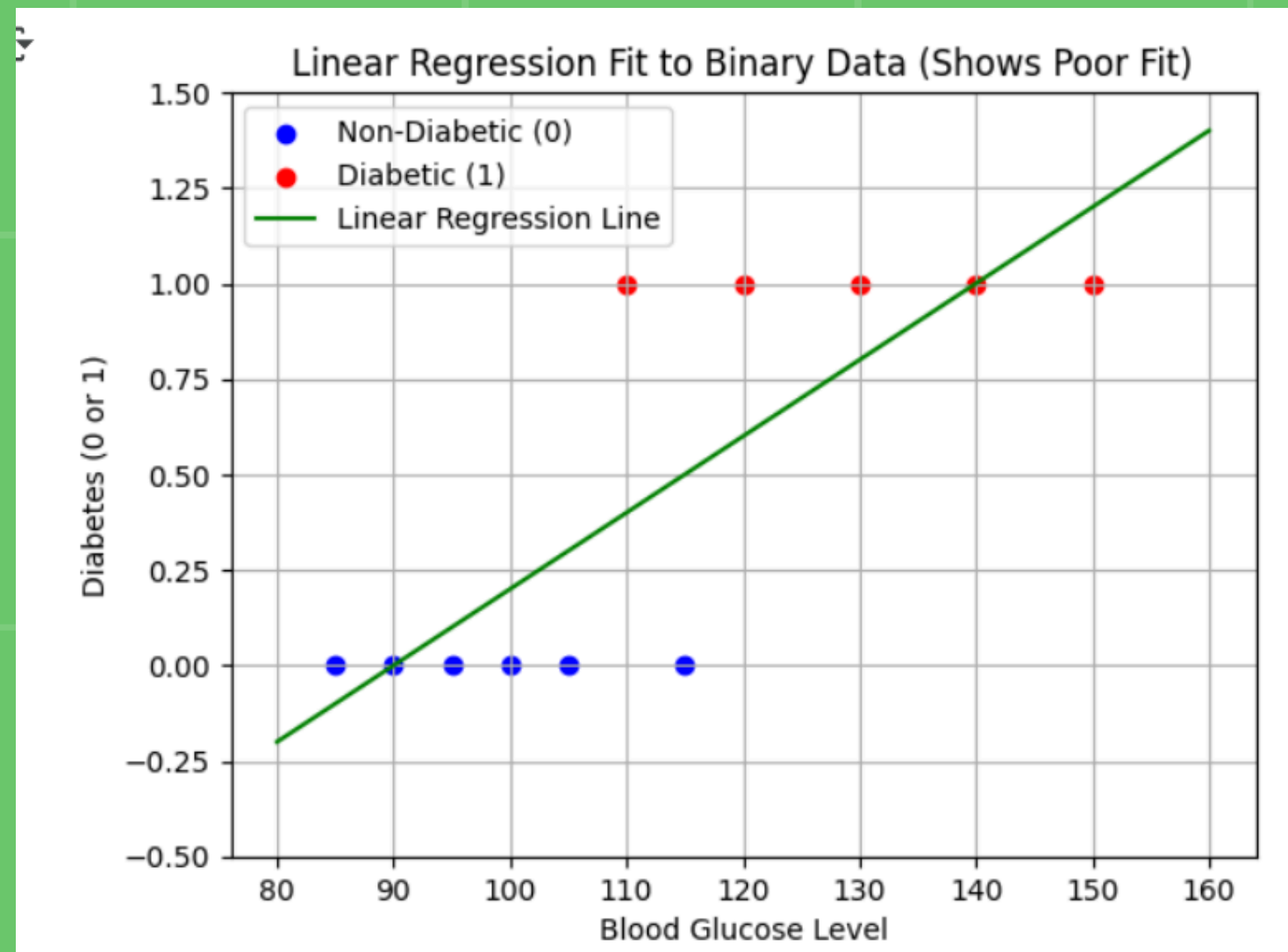
REGRESSION

Linear Regression vs Logistic

	Linear Regression	Logistic Regression
Type	Regression	Classification
Purpose	Predict continuous numeric values	Perform binary classification
Output	Continuous value (any real number)	0 or 1
Output Range	$[-\infty, +\infty]$	$[0, 1]$

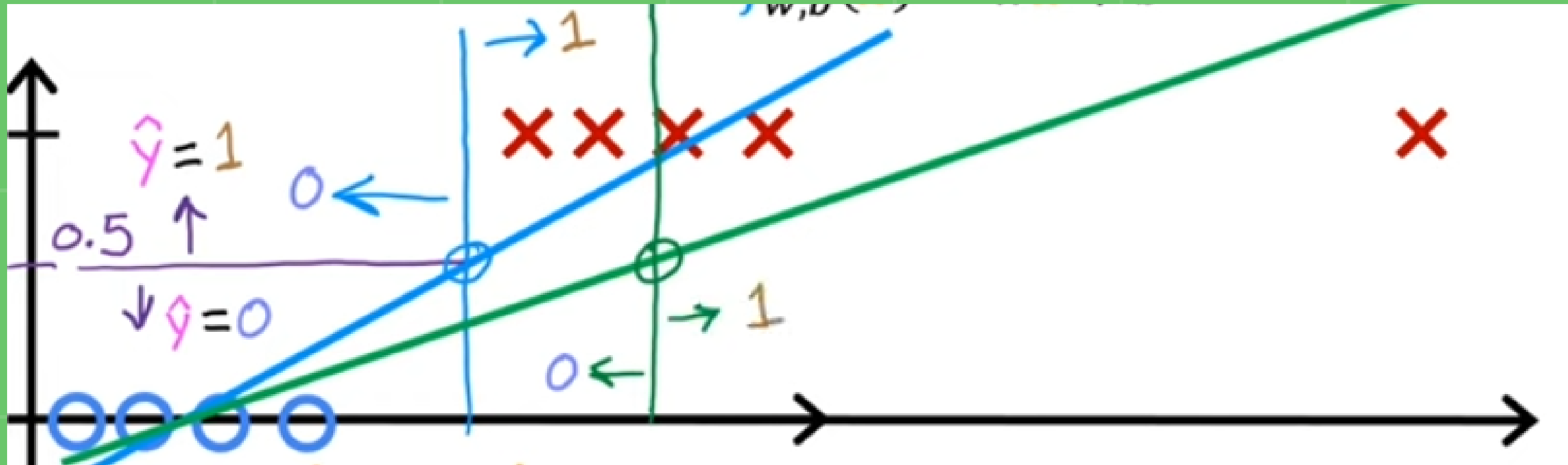
Why We Can't use Linear Regression?

- **Problem with Continuous Output:** Linear regression predicts continuous values like 5.7, and our target are discrete numbers, 0 and 1
- If we want to set a threshold, how to set a one when the output range is $[-\infty, +\infty]$?



Why We Can't use Linear Regression?

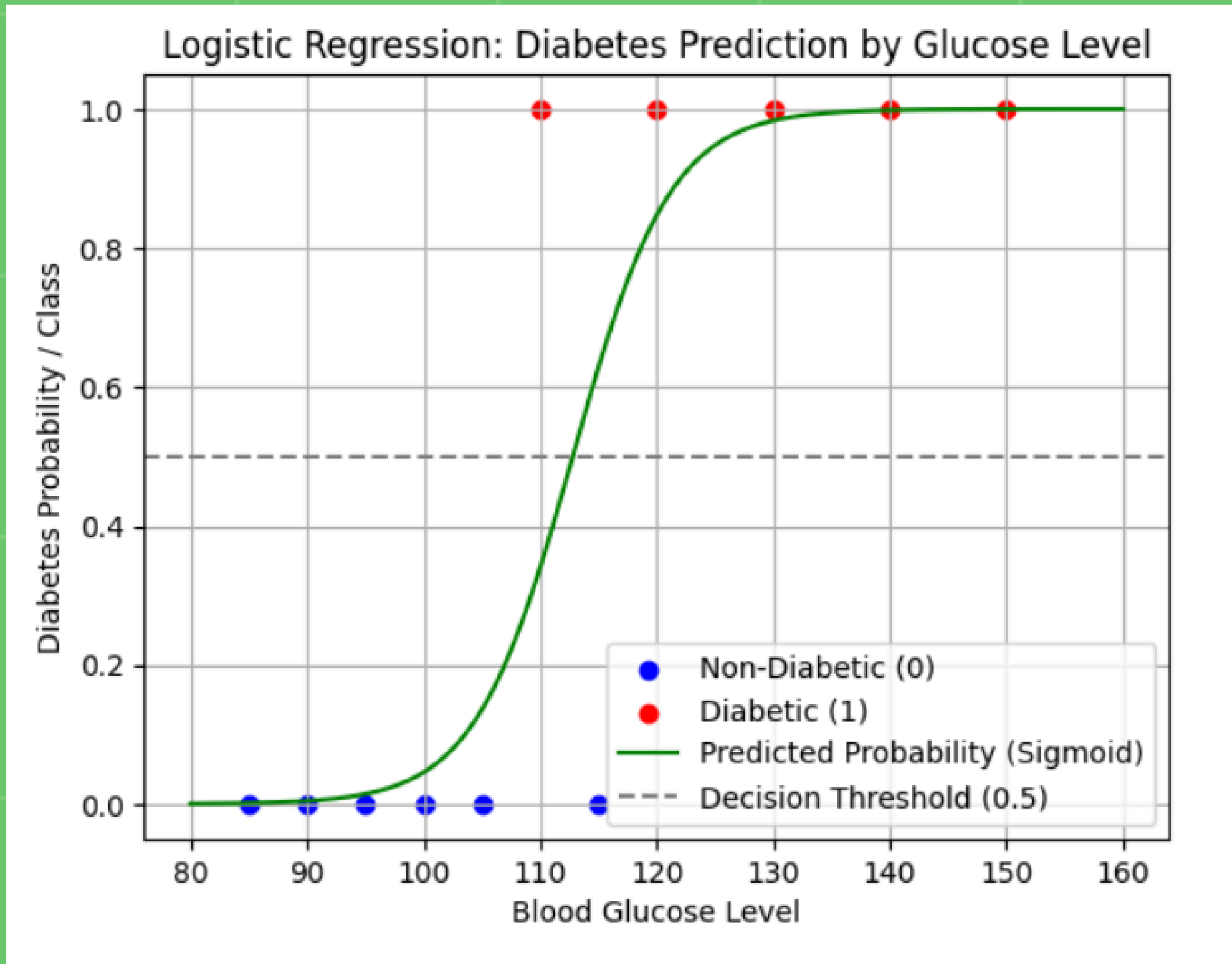
- **Problem with outliers and the Decision Boundary:** Linear regression is sensitive to outliers. extreme points can skew the line, leading to misclassification.



Solution

- **We predict probabilities** (from 0 to 1): how likely is this person to be diabetic?
- **We set a threshold (commonly 0.5)** to convert probabilities into class predictions:
 - = If probability > 0.5 → predict diabetic (1)
 - = Else → predict non-diabetic (0)
- Instead of a linear equation that can output any real number $(-\infty, +\infty)$, logistic regression uses a **sigmoid function**, which is bounded between 0 and 1.

Sigmoid Function





Logistic Regression

How Logistic Regression Works?

1- Compute the Linear Combination (z): We start by calculating a linear equation that combines the input features.

Each feature is multiplied by a weight that the model will learn, and we also add a bias term.

$$z = w \cdot x + b$$

How Logistic Regression Works?

2- Apply the Sigmoid Function to Get Probability: The value of z can be any number from $-\infty$ to ∞ , but we want a probability between 0 and 1. So, we use the sigmoid function:

$$y = \sigma(z) = \frac{1}{1 + e^{-z}} = \frac{1}{1 + \exp(-z)}$$

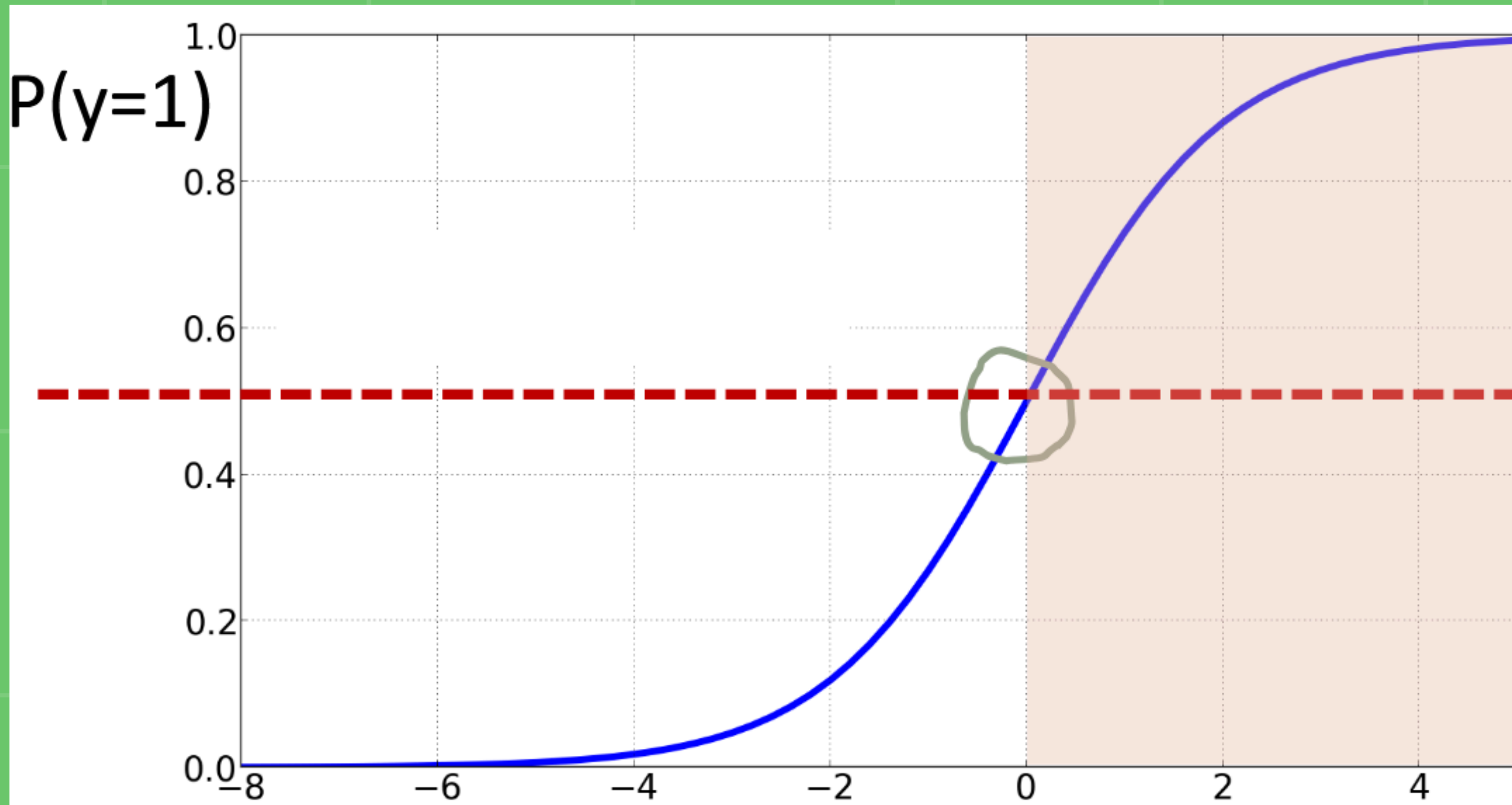
How Logistic Regression Works?

3- Make a Decision with a Threshold: Once we have the probability, we can make a final prediction:

$$\hat{y} = \begin{cases} 1 & \text{if } P(y = 1|x) > 0.5 \\ 0 & \text{otherwise} \end{cases}$$

0.5 here is called a decision boundary

How Logistic Regression Works?



How Logistic Regression Works?

4- Measure Error Using Cross-Entropy Loss: To train the model, we need to measure how wrong the predictions are. Since we're predicting probabilities, we use a special loss function called cross-entropy.

$$\text{Loss} = - (y \cdot \log(\hat{y}) + (1 - y) \cdot \log(1 - \hat{y}))$$

How Logistic Regression Works?

5- Update the Weights with Gradient Descent: The model tries to find the best weights $w_1, w_2, \dots, w_n$ and bias b that minimize the total loss.

Gradient descent algorithm

repeat until convergence {

$$w = w - \alpha \frac{\partial}{\partial w} J(w, b)$$

$$b = b - \alpha \frac{\partial}{\partial b} J(w, b)$$

}

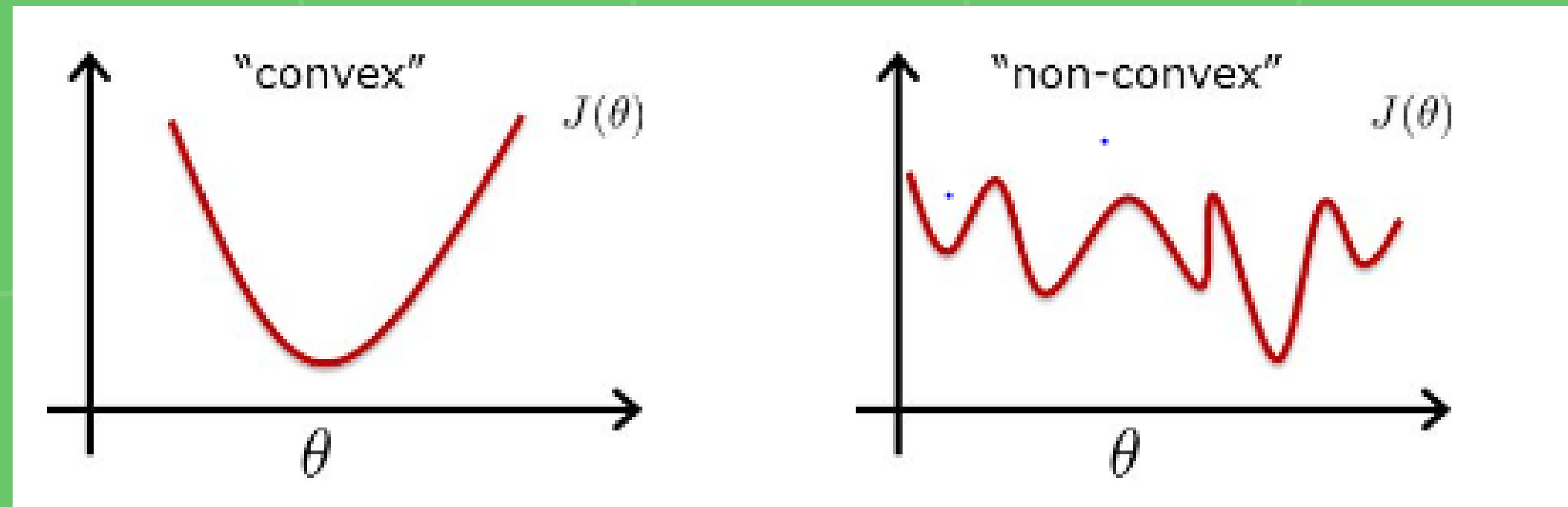


Cost Function

Why not use MSE?

1- Mismatch in Assumptions: MSE assumes that the output is a continuous value, but in logistic regression, the output is a probability between 0 and 1

2- Non-convex surface: Sigmoid function is non-linear, and when we use MSE with it, the resulting loss function is non-convex that have many global minima and hard to converge

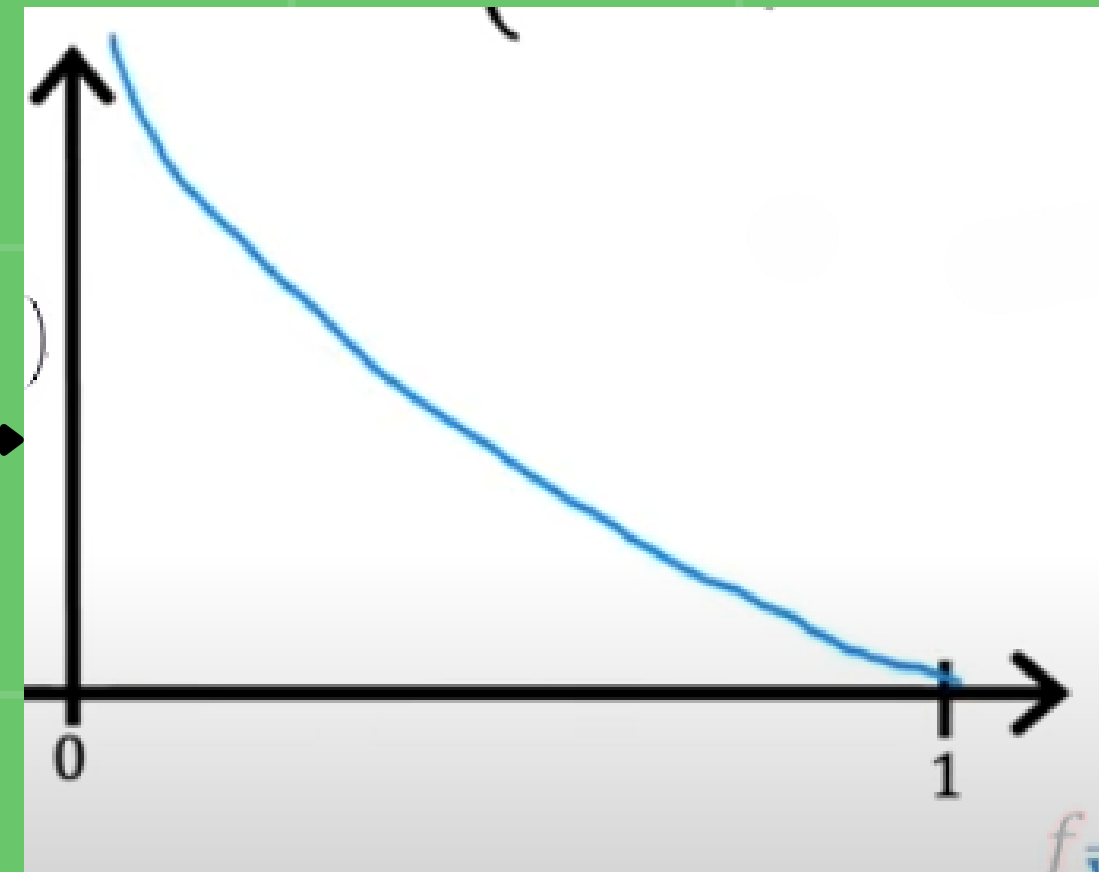
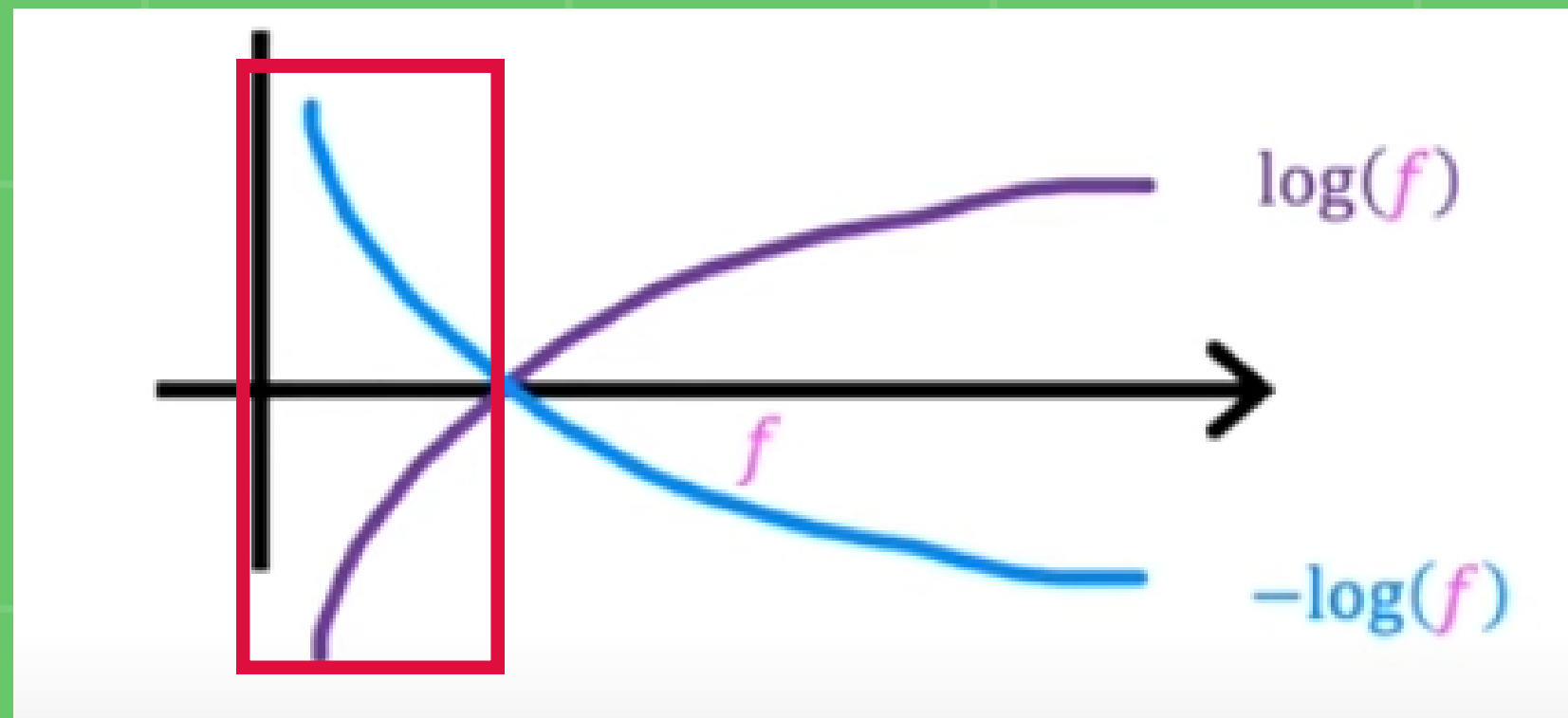


Log loss (Cross-Entropy)

- In logistic regression we use a loss function called log loss or cross entropy.

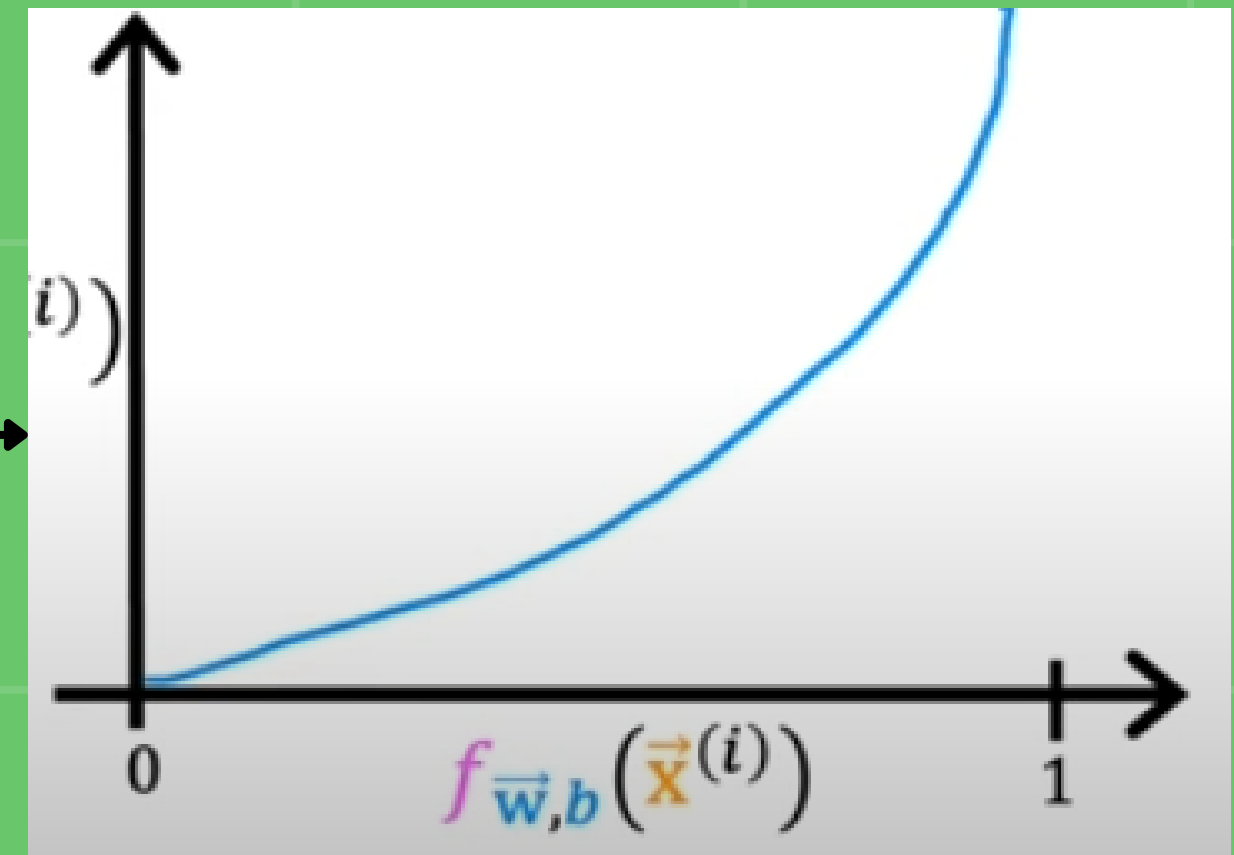
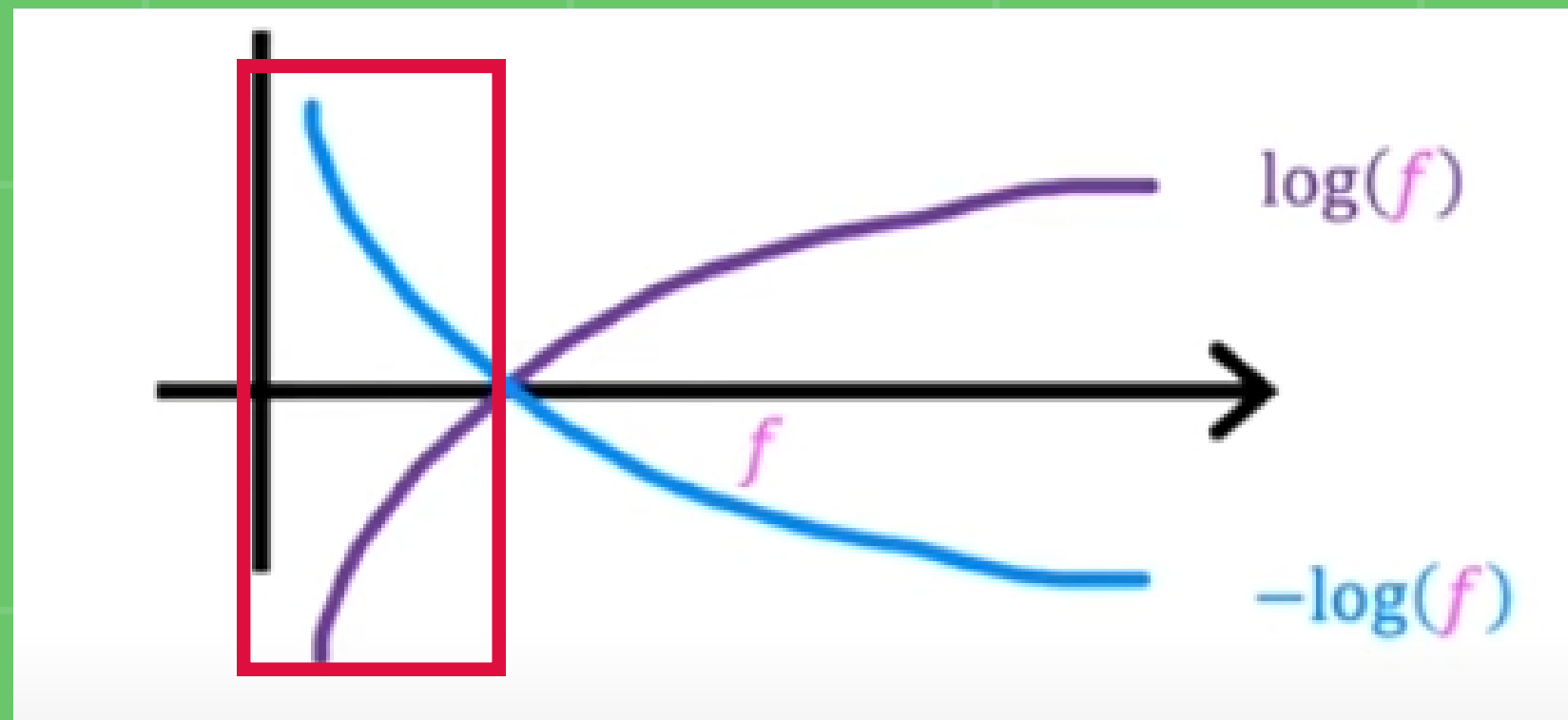
$$L(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)}) = \begin{cases} -\log(f_{\vec{w},b}(\vec{x}^{(i)})) & \text{if } y^{(i)} = 1 \\ -\log(1 - f_{\vec{w},b}(\vec{x}^{(i)})) & \text{if } y^{(i)} = 0 \end{cases}$$

Log loss (Cross-Entropy)



If the actual label = 1 and predicted label is close to 1 then the loss will be a small number and vice versa

Log loss (Cross-Entropy)



If the actual label = 0 and predicted label is close to 0 then the loss will be a small number and vice versa

Log loss (Cross-Entropy)

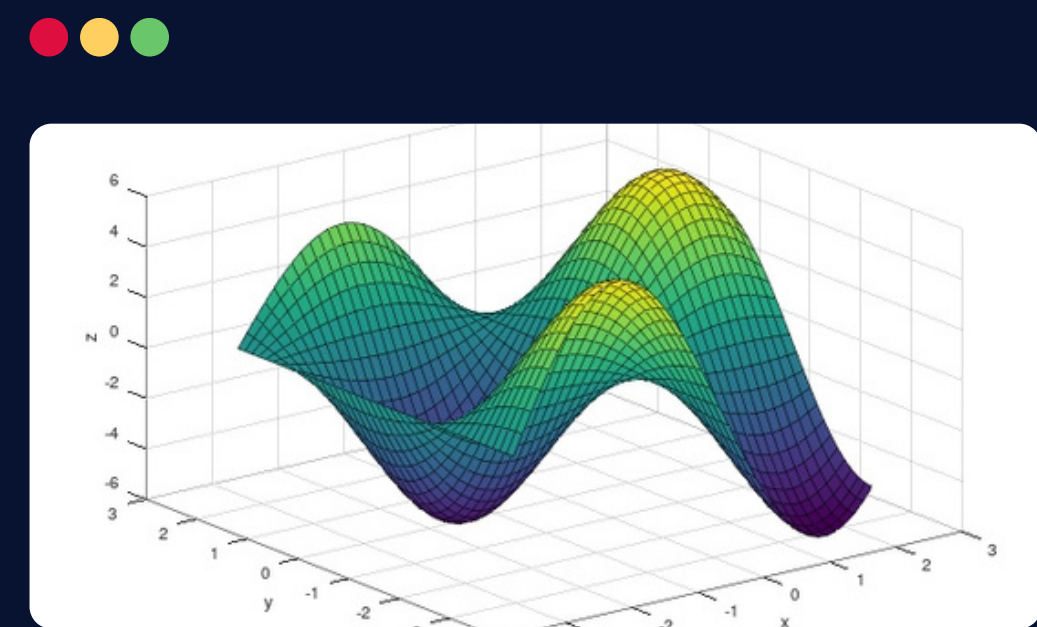
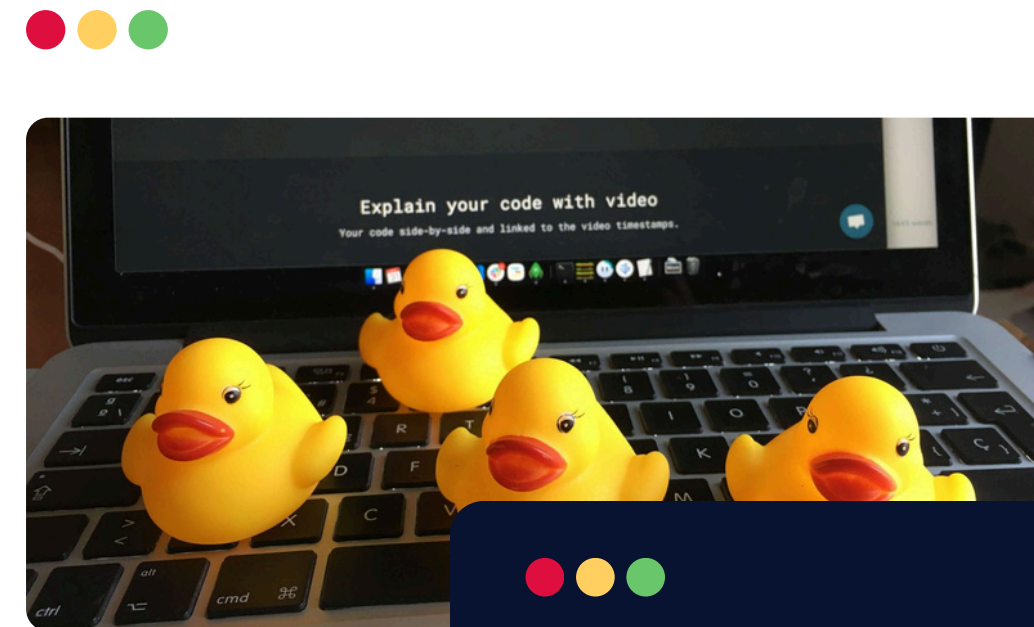
$$J(w) = -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)}) \right]$$

- if the actual value $(y^i) = 0$, then we get

$$(1 - y^{(i)}) \log(1 - \hat{y}^{(i)})$$

- if the actual value $(y^i) = 1$, then we get

$$y^{(i)} \log(\hat{y}^{(i)})$$



Thank You

created by:
Ahd Ehab